

F24-171-D-PenTrack

Project Team

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Chapter 1

Introduction

The purpose of this document is to propose the development of a system that analyzes a football/soccer striker's body pose during penalty kicks. It would predict the outcomes of the shot and simulate a goalkeeper's reaction. The goal is to help players improve their penalty kicks. Penalties are a crucial aspect of the game. They often become the determining factor of matches. Players, especially in places like Pakistan where football isn't as popular, find it difficult to practice their shots against tough goalkeepers. Hence the project aims to provide a platform to bridge that gap. This would be done by using pose estimation and machine learning to analyze a striker's body movements, predict shot outcomes, and simulate realistic goalkeeper reactions.

1.1 Problem Statement

The project would allow football players to enhance their penalty kick performance. By providing insights into how their pose affects shot direction and power a player would optimize their performance.

Players find it challenging to understand how body movements influence the outcome of their shots. Traditional coaching may not provide feedback on body positioning. This makes it difficult for players to refine their technique. Hence, the project aims to solve this problem by analyzing body poses, predicting shot results, and simulating goalkeeper reactions.

In summary, the lack of precise pose-based insights during penalty kick training results in capped performance. This project approaches this gap by developing a model that links body pose to shot outcomes and provides a simulation environment. This would ultimately help players to master penalty kick techniques.

1.2 Scope

The scope of the project is to develop software that would help football players enhance their penalties. It would do this by analyzing body pose during the kick and the dummy keepers' reaction. The main functionalities of the system include pose estimation, shot prediction, and goalkeeper simulation. Pose estimation tools like OpenPose[1] or MediaPipe[2] will be used to extract key body points from the penalty footage. After that the spatial features will be passed by an LSTM network to understand the sequence of movements occurring during a penalty kick. The model would predict outcomes such as direction and power of the shot. The direction would be classified as top left, top right, bottom left, etc. Also some might wonder why not use the data of direction and power from the users footage, well for it to be realistic the model should get the same amount of information as what a keeper would get during a penalty. That amount of information is of a very short time-frame in which a keeper only registers information till the player strikes the ball.

The proposed system would also include a simulation in which a dummy keeper would react to the predicted shot outcome. Based on the balls trajectory and speed, the goalkeepers reaction would be classified into certain regions such as bottom left, bottom right, top mid etc. For example, if the model predicts the ball to go to the bottom right, then a pre-trained animation of a goalkeeper would perform a saving action towards that direction. This simulation would provide a close to realistic training environment. This would enable players to make adjustments helping them improve their performance. The proposed software would also include a user-friendly interface where players and coaches would be able to upload videos and analyze the results. The system would prove to be a valuable tool for players and coaches, especially in regions where access to high-level training and keepers are limited.

Overall, the project's scope highlights the development of a platform that would help bridge the gap between traditional training and advanced technological analysis. Hence, this will contribute to more effective and targeted practice sessions.

1.3 Modules

The project is divided into several modules, like Data Collection, Model Training, Mobile App Development and Simulation. Each module focuses on specific functionalities and features which play a key role in the overall project.

1.3.1 Data Collection

[In this module we will focus on dataset collection to train our model]

1. Data Collection (YouTube, SoccerNet^[3])
2. Data Cleaning and Preprocessing
3. Partial Annotation
4. Feature Extraction

1.3.2 Model Training

[In this module we will focus on training our selected models on the training dataset]

1. Pose Data Processing and Feature Engineering
2. Model Training and Fine-Tuning
3. Output Validation and Performance Evaluation

1.3.3 App Development

[In this module we will focus on the mobile app development phase in which we will simulate our model results]

1. UI/UX
2. Application Development
3. Application Integration (integrating models/simulation)
4. Testing and Validation

1.3.4 Simulation

[In this module we will simulate the model results in a 3D environment]

1. **Setting up the 3D Environment**
 - Setup a 3D space to showcase the animations.

- Load the pre-stored goalkeeper animations matching the appropriate shot directions.

2. **Rendering Final Results**

- Based on the model's prediction, generate an appropriate animation to visualize the goalkeeper's reaction.
- Display the animation after processing users footage, aligning with the predicted ball trajectory to show how the goalkeeper would attempt to make a save.

1.4 User Classes and Characteristics

Identify the various user classes that you anticipate will use this product, and describe their pertinent characteristics.

User class	Description
Player	A player is a individual who wants to improve his penalty kicks, in order to be proficient during penalty shootouts. Players can range from beginners to professionals and can improve their penalty kicks by analyzing their body poses and shot outcomes. The players can use the app individually or under supervision of trainers.
Coaches	Coaches are individuals who provide individual training to players in order to improve their performance. Coaches can use the app to analyze body posture and shot outcomes, then give personalized feedback to players to improve their penalty kicks.
Football Clubs	Football clubs can use the app in order to standardize their training mechanism for all players, especially when it comes to penalties. Each club consists of multiple players, and to maintain consistency in training, clubs need to manage player training in a centralized manner. This will be tasked to Coaches employed by the club. Furthermore, clubs can keep a track of overall improvements in penalty performance by analyzing player performance analytics.
Researchers	Researchers are individuals who can bring further innovation to sports analysis tools by bringing enhancements to existing systems. They can use existing app to learn existing workflow and provide further optimization and improvements to the existing tools in order to further improve player performance.

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Event Response Table

- Event- response table for our app in which most of the functionalities are performed at backend. The following table highlights specific events and the corresponding response for those events with the time constraints.

Event	Response	Time Constraint
Player wants to upload a penalty kick video	System validates the player's login status, then confirms the player's navigation to the upload section.	Immediate (within 1 second)
Player selects a video from gallery or records a video to upload	System validates the file format and size.	Within 5 seconds
Player selects a valid video	System uploads the video and notifies user of successful upload.	Within 20 seconds
Player selects an invalid file format	System displays an error message and prompts for a valid file.	Within 3 seconds
Upload failure occurs	System notifies the player of upload failure and prompts user to retry.	Within 3 seconds (after failure)
Player wants to view penalty kick analysis results	System verifies if analysis results are generated.	Within 5 seconds
Results are available	System retrieves and displays shot outcome and goalkeeper reaction for user's uploaded video.	Within 20 seconds
Results are not available	System notifies player of ongoing analysis and prompts to retry later.	Within 3 Seconds
Processing error occurs	System notifies player of processing failure and suggests retry.	Within 3 Seconds
Player requests to view their past performance statistics	System verifies player login and data availability.	Immediate (within 1 second)
Past data is available	System retrieves and displays statistics such as goals scored and shot power.	Within 5 seconds
No statistics available	System notifies player that data is unavailable.	Within 3 Seconds
Data retrieval error occurs	System displays error and asks user to retry.	Within 3 Seconds
Player chooses to share penalty analysis results	System verifies if the results are available.	Immediate (within 1 second)
Results are available	System generates and provides shareable content (link or post).	Within 10 seconds
No results available	System notifies player that results are not available yet.	Immediate (within 1 second)
Sharing failure occurs	System notifies player of sharing failure and suggests retry.	Immediate (within 1 second)

Figure 2.1: Event Response Table

2.2 Functional Requirements

This section describes the functional requirements of the system expressed in the natural language style broken down into different app modules/components.

2.2.1 Video Upload & Preprocessing

Following are the requirements for the Video Upload & Preprocessing module:

1. The app shall allow users to upload their recorded penalty kick videos.
2. The app shall do preprocessing on the uploaded videos to extract relevant frames

for analysis.

3. The app shall validate the uploaded video format and size to ensure compatibility.

2.2.2 Pose Estimation Module

Following are the requirements for the Pose Estimation Module:

1. The app shall extract body pose keypoints from the video frames.
2. The app shall store the extracted keypoints for subsequent processing.

2.2.3 Prediction Module

Following are the requirements for the Prediction Module:

1. The app shall process video data to make predictions about shot outcomes (shot direction and power).
2. The app shall validate the predictions to ensure consistent results.

2.2.4 Simulation Module

Following are the requirements for the Simulation Module:

1. The app shall render simulation results based on the model's predictions.
2. The app shall simulate the interaction between the penalty shot and a goalkeeper model reacting to the predicted outcome.
3. The app shall provide a visual representation of the goalkeeper's reaction alongside the simulation results.

2.2.5 App Module

Following are the requirements for the App Module:

1. The app shall provide a user-friendly interface for users to navigate through the app.
2. The app shall allow users to easily upload videos and view simulation results.
3. The app shall ensure that the insights provided on penalty kicks are easy to understand for non-technical users.

2.3 Non-Functional Requirements

This section specifies nonfunctional requirements. These quality requirements should be specific, quantitative, and verifiable.

2.3.1 Reliability

The app shall ensure that the prediction results for different videos uploaded by players exhibit a consistency rate of at least 95% across multiple tests under varying conditions.

The app shall guarantee that data is saved reliably, with a data retention rate of 99.9%, ensuring that no more than 0.1% of data is lost during storage or processing.

The app shall generate accurate simulations, achieving an accuracy rate of at least 80% when comparing predicted outcomes to actual outcomes based on historical data.

2.3.2 Usability

At least 90% of users shall complete key tasks (e.g., uploading a video, viewing results) without assistance during usability testing, indicating intuitive navigation.

The insights provided on penalty kicks shall be rated as "easy to understand" or higher by at least 80% of users in post-interaction surveys.

2.3.3 Performance

The model should be efficient enough to provide the penalty results within (10-20) seconds for a player's uploaded video. In addition, users should be able to upload their videos successfully within 20 seconds. Moreover, the simulation should also run smoothly on at least 85% devices without buffering or lag.

2.3.4 Security

User's personal data (like their uploaded videos/ insights) should be stored securely and made sure to only be made visible to authenticated users. Moreover, the app should ensure no unnecessary data is saved for enhanced security.

2.4 Domain Model

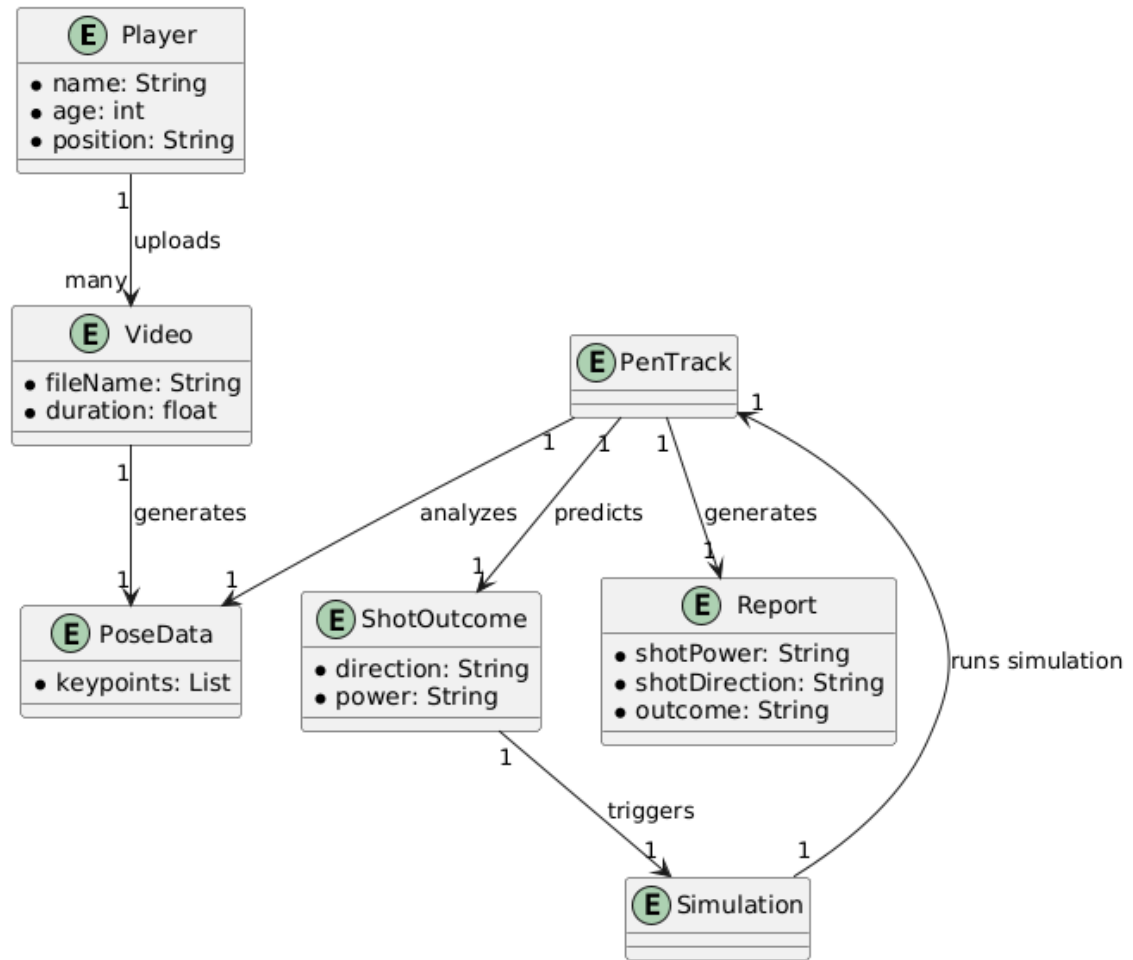


Figure 2.2: Domain model

Chapter 3

System Overview

Give a general description of the functionality, context, and design of your project. Provide any background information if necessary.

3.1 Architectural Design

The system will be divided into several interconnected modules that work together to achieve the functionality of predicting penalty shot outcomes and simulating goalkeeper reactions based on body pose analysis. The primary modules include:

1. **Video Upload & Preprocessing:** This module handles video uploads from users and preprocesses the footage to extract relevant frames.
2. **Pose Estimation Module:** Uses MediaPipe to extract body pose keypoints from the video frames.
3. **Prediction Module:** This module contains the machine learning model trained to predict shot outcomes based on pose data.
4. **Simulation Module:** Simulates the interaction between the penalty shot and a goalkeeper model that reacts based on the predicted outcome.
5. **UI/UX Module:** Provides a user interface for players and coaches to upload videos, view simulation results, and interact with the simulation environment.

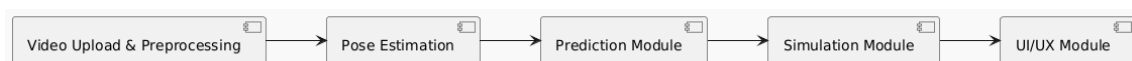


Figure 3.1: A Box and Line diagram.

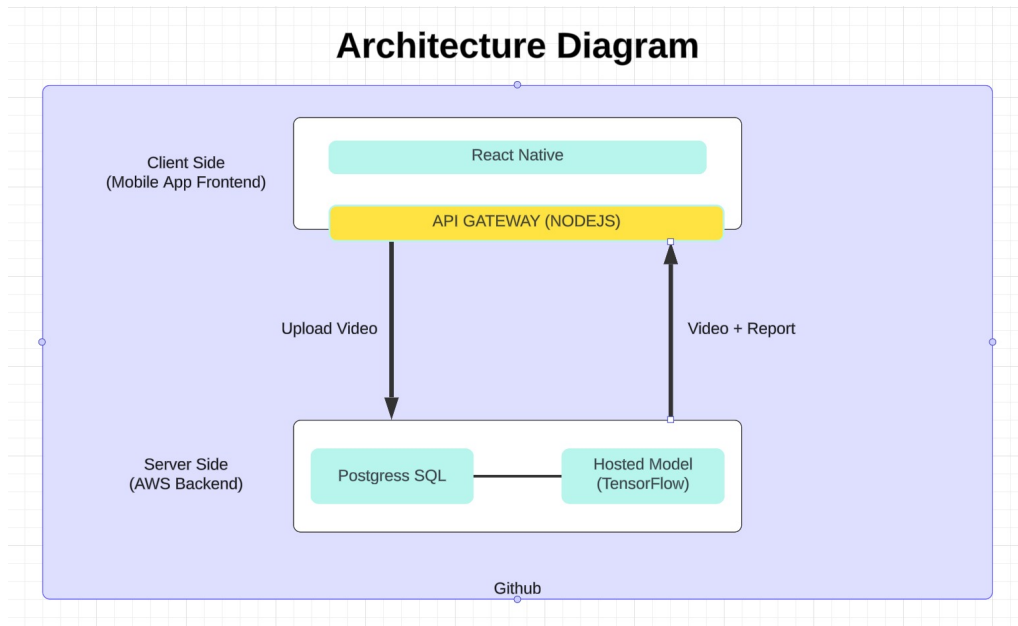


Figure 3.2: Architecture Diagram

Architecture Description: The architecture diagram illustrates the key components of PenTrack app, highlighting the client-server architecture which is ideal for apps that require real-time data in order to fetch updates from a server.

- **Client Side (Frontend):** The mobile application, built using React Native, serves as the interface for users to interact with the system, enabling video uploads and the viewing of results.
- **API Gateway:** Node.js acts as an intermediary, routing requests between the mobile app and backend services, ensuring smooth communication.
- **Backend Services (AWS):** This includes a PostgreSQL database for storing user data and video, while the TensorFlow model processes the uploaded videos to provide predictive analytics.
- **Data Flow:** A User upload's their video, which is stored in the database and are then processed by the machine learning model, and results are sent back to the mobile app and displayed to the user.
- **Version Control:** Integration with GitHub for version control and code repository management. This allows for collaborative development between team members, tracking changes, and maintaining the codebase effectively.

3.2 Design Models

1. Activity Diagram:

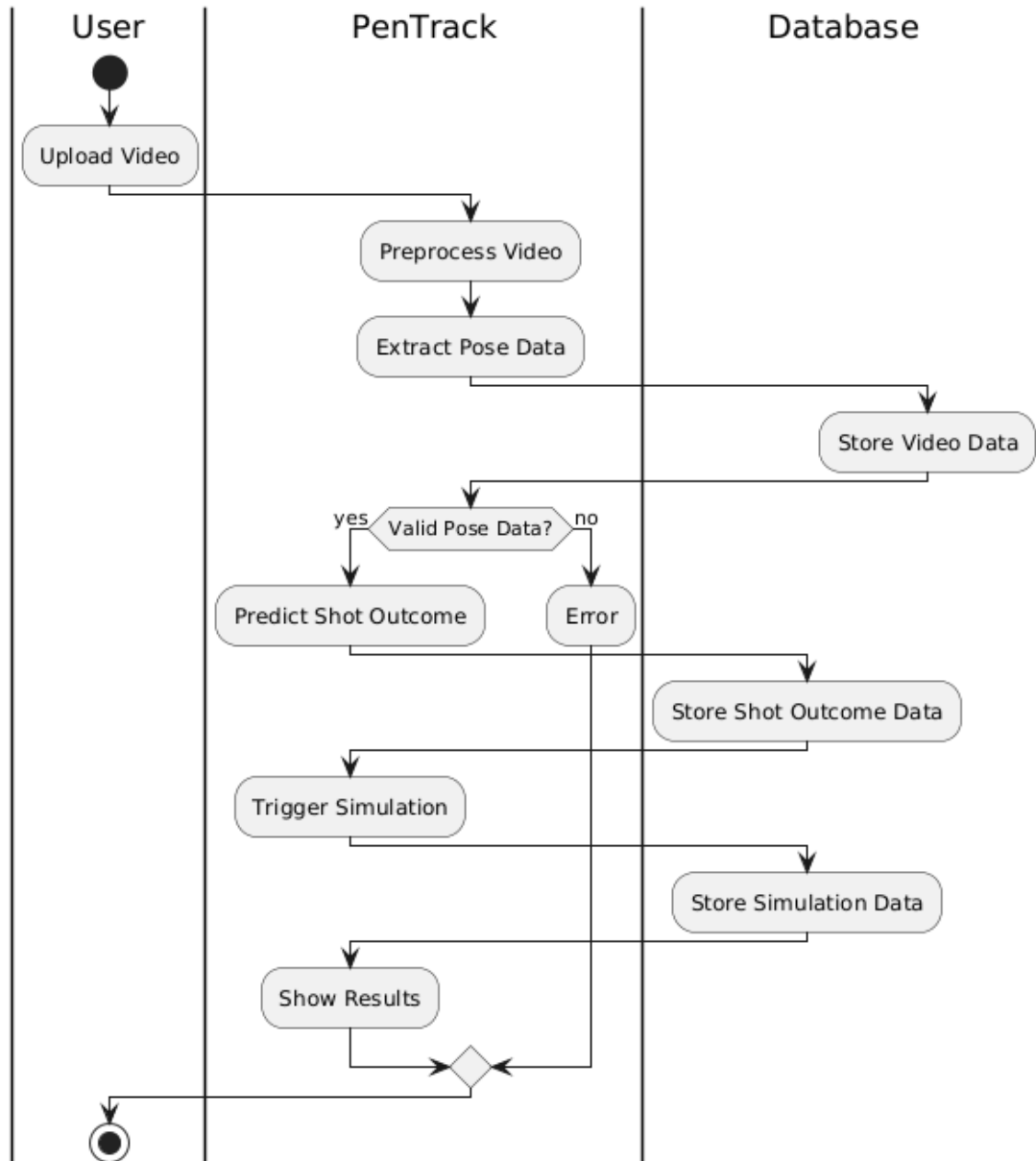


Figure 3.3: Activity diagram.

2. Class Diagram:

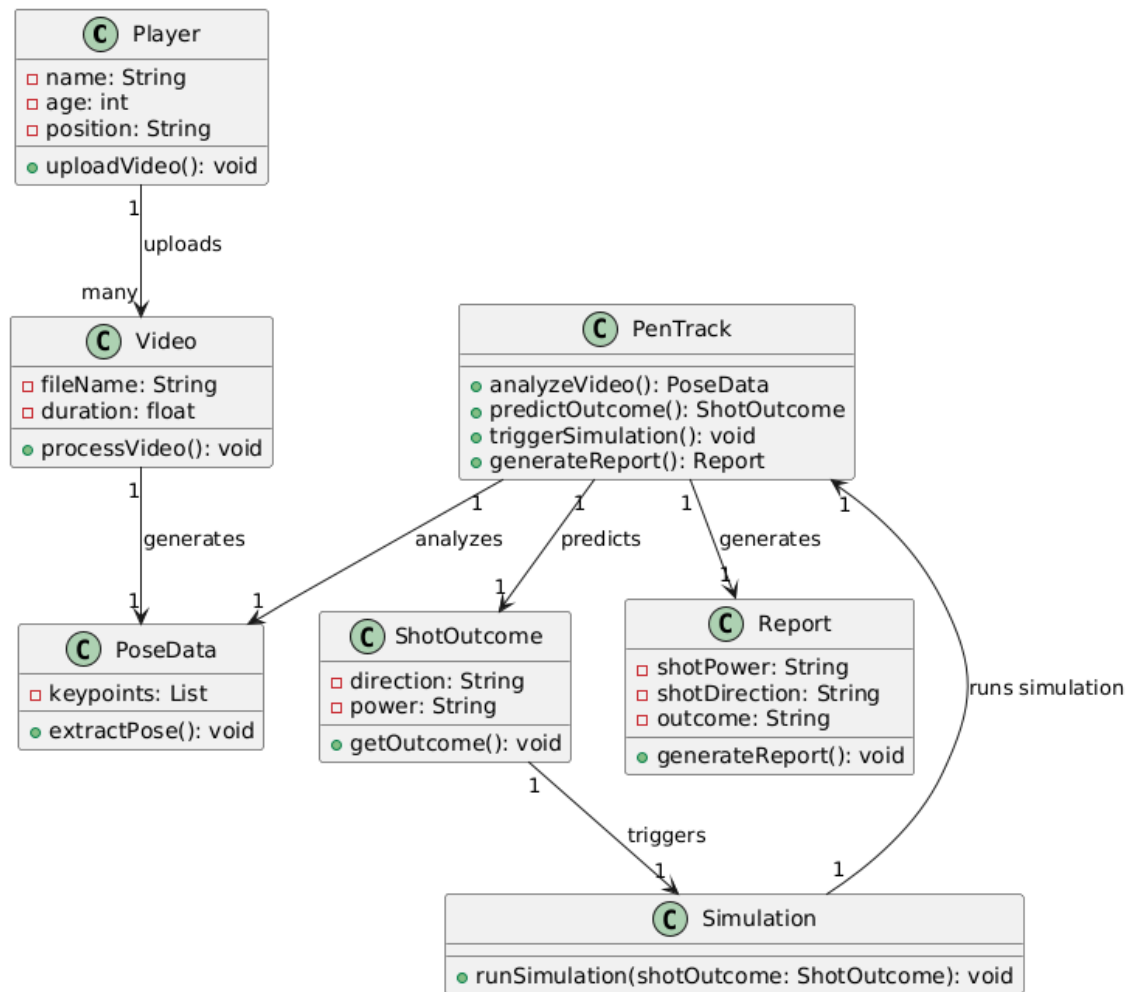


Figure 3.4: A Class diagram.

3. Class-level Sequence Diagram:

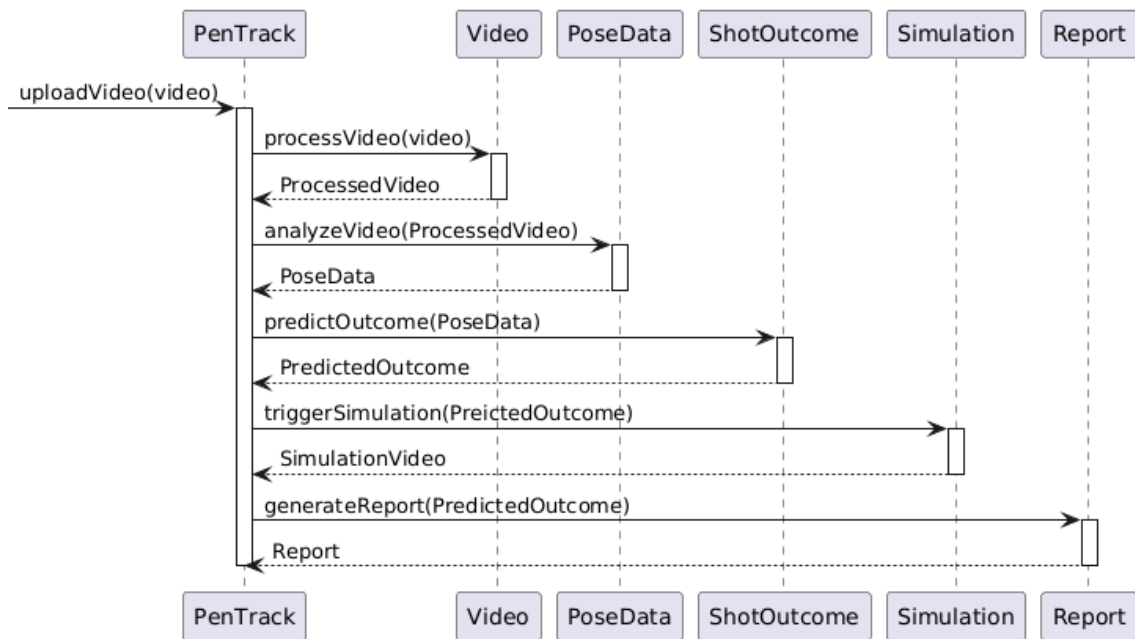


Figure 3.5: Class-level Sequence diagram.

4. State Transition Diagram:

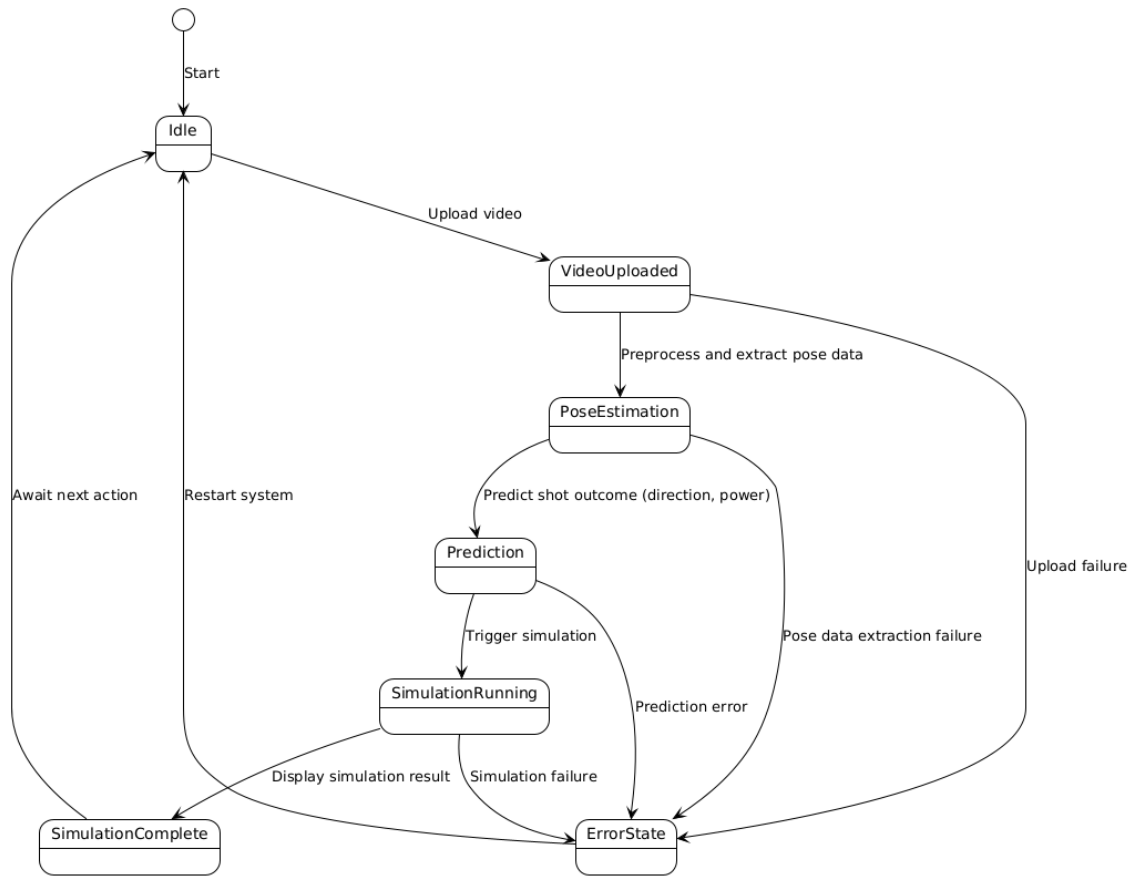


Figure 3.6: State Transition diagram.

5. ERD: Diagram Description

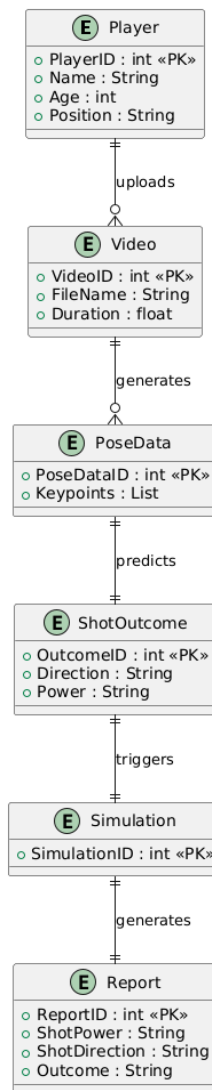


Figure 3.7: Entity Relation Diagram

Entities: The diagram includes entities such as Player, Video, PoseData, ShotOutcome, Simulation, and Report, each with their respective attributes and primary keys.

- The relationship from **Player** to **Video** indicates that a player can upload multiple videos.
- The relationship from **Video** to **PoseData** shows that each video generates multiple pose data entries.
- The relationships to **ShotOutcome**, **Simulation**, and **Report** indicate that each entry progresses through the process of predicting outcomes, triggering simulations, and generating reports.

3.3 Data Design

In the **Data Design** section, we will outline how data will be handled and stored within the system.

Data Structures:

1. **Pose Data:** This consists of 3D coordinates (x, y, z) for each key body point extracted from videos.
2. **Shot Outcome Data:** Stores shot characteristics like direction (top left, bottom right, etc.) and power (low, medium, high).
3. **Video Data:** Raw videos uploaded by users are stored temporarily for processing, with extracted pose data stored for long-term usage.
4. **Simulation Data:** Stores the results of each simulation, including the goalkeeper's movements and the outcome of the penalty.

Storage:

- **Database:** The system will use a relational database (e.g., PostgreSQL) to store user information, processed pose data, and simulation results.
- **File Storage:** Videos will be stored in a cloud-based solution or file server for fast retrieval during analysis and simulation.

Bibliography

- [1] Zhe Cao, Gines Hidalgo, Tomas Simon, Shih-En Wei, and Yaser Sheikh. Openpose: Realtime multi-person 2d pose estimation using part affinity fields, 2019.
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- [3] SoccerNet. Soccernet: A scalable dataset for action spotting in soccer videos, 2024.